The Generative Time Circuit Theorem

Complete Proofs, Derivations, and Applications

Principia Orthogona Working Paper GTCT-2026-001

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Abstract

We prove the Generative Time Circuit Theorem (GTCT, Theorem T1): in the TOGT operator algebra on a contact manifold, time is the generative circuit operator T -- the fifth stage of the operator chain C -> K -> F -> U -> T -> source. The g-series is a taxonomy of behavioral regimes indexed by the labels g^0, g^2, g^6, g^33, g^64; the indices are motivated labels, not derived constants. For any dm3 system that has completed at least g^33 = 33 generative cycles (the heuristic stability index, stated precisely as a conjecture), the return x_0 -> G^64(x_0) -> G^64(x_64) -> x_0' with x_0' != x_0 is a spiral, not a loop: the source is enriched by one completed circuit. The action variable z accumulates dissipation along the return, encoding the thermodynamic cost of the full circuit in the contact form alpha = dz - lambda. Structural stability within the dm3 radius epsilon_0 = 1/3 (Theorem D, Volume II) is proved. Connections to quantum delayed-choice experiments, the two-state vector formalism, and entanglement swapping are presented as open research problems, not theorems. Three independent falsifiability conditions are provided. Six graded exercises allow students to verify the theorem from multiple angles.

Keywords: generative operator chain; contact geometry; dm3; time operator; spiral return; stability threshold; Principia Orthogona; GTCT

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1. Where This Theorem Lives in the Series

The Principia Orthogona series establishes a five-stage operator chain C -> K -> F -> U -> T governing generative transitions on contact manifolds. The first four operators -- Compression, Curvature constraint, Fold, Unfolding -- are established in Volumes I and II (Nogueira Grossi 2025a, 2025b) and verified in Lean 4 with zero axioms beyond Mathlib4 in the AXLE repository (github.com/TOTOGT/AXLE). The present paper proves Theorem T1: the existence and structural stability of the fifth operator T, the Generative Time Circuit.

Volume / Ring	Content
Volume I / Ring 1	C -> K -> F -> U on Riemannian manifold. Theorems A-D. Zenodo 19117400.
Volume II / Ring 2	dm3 contact mechanics. epsilon_0 = 1/3. Canonical invariants (2pi, -2, 2). Zenodo 19379385.
Volume III / Ring 3	Biological instantiations: HPA, neural, circadian, immune. Zenodo 19208015.

AXLE / Ring 4	Lean 4 formal verification engine. github.com/TOTOGT/AXLE.
This paper / Ring 5	**GTCT: the time circuit closes. T is the fifth operator. Spiral return proved.**

2. The g-Series: A Regime Taxonomy

The dm3-GTCT framework uses a taxonomy of behavioral scales called the g-series. This section defines the taxonomy precisely. The g-series makes no mathematical claims about the index values; the indices are labels chosen by the criteria stated below, not derived constants.

Definition 2.1 (g-Series Taxonomy).

The g-series is a finite ordered set of regime labels:

```
g^0 < g^2 < g^6 < g^33 < g^64
```

Each label g^k names a qualitative behavioral regime of a dm3 system. The superscript k is an ordinal index, not an exponent. The series is not a sequence of numbers; it is a classification.

Definition 2.2 (Index Criteria).

The five indices are assigned by the following criteria, each corresponding to a qualitatively distinct stability property of the operator chain G = U * F * K * C:

Label	Index	Regime Name	Criterion
g^0	0	Seed	Single operator application; no closure condition.
g^2	2	Compositional	Two-operator closure; minimal self-reference (F active).
g^6	6	Cycle-scale	Full G-cycle closure; limit cycle Gamma entered.
g^33	33	Soft equilibrium	Heuristic stability index: ceil(log2(3!) * 4) * 3 = 33. See Remark 2.3.
g^64	64	Phase threshold	Complete possibility-space coverage; 2^6 = 64 operator-state combinations.

Remark 2.3 (Status of the index 33).

*The index 33 arises from the following heuristic computation: there are three invariants that must close simultaneously (orthogonality, nilpotency of deviations, spectral collapse), giving 3! = 6 orderings; log2(6) * 4 is approximately 10.34, so ceil(log2(3!) * 4) = 11; multiplying by 3 for a triple-confirmation robustness criterion gives 3 * 11 = 33. This computation is heuristic. The 'multiply by 3' step reflects a design choice -- that three independent closures are required for robust stability -- which is not derived from the contact-geometric structure of Sections 5-6. The index 33 is therefore a motivated label, not a theorem. Its mathematical status is that of a conjecture: we conjecture that dm3 systems require at least 33 operator cycles for the three invariants to achieve simultaneous robust closure, and we identify this as an open verification target for the AXLE Lean 4 repository.*

Remark 2.4 (Crystal notation).

The notation g6:33 and g33:33 used elsewhere in the Principia Orthogona series denotes association, not equality. 'g6:33' means: the cycle-scale regime g^6 carries the structural index 33. It is a descriptor of the same kind as 'D6 symmetry: 6-fold' or 'E8 lattice: rank 8'. No algebraic identity g^6 = 33 is claimed.

3. Definitions and Axioms

Definition 3.1 -- Generative Operator Chain.

The single-cycle generative operator is G = U o F o K o C, where:

C (Compression): reduces complexity to a minimal viable seed; injective and contractive.

K (Constraint): defines the curvature boundaries that make form possible; decreases potential Phi.

F (Fold): creates the first self-reference, curving the system back onto itself; non-injective with finite branch set.

U (Unfolding): releases latent potential into visible reality; decreases Phi and drives orbits toward a fixed point.

One complete application of G is one generative cycle. The full chain including the fifth operator is:

```
C -> K -> F -> U -> T -> source
```

Definition 3.2 -- Stability Threshold g^33 = 33.

The index 33 is the heuristic stability threshold for regenerative coherence -- simultaneous closure of the three invariants (orthogonality, nilpotency of deviations, spectral collapse). The computation is:

```
n_min = ceil( log2(3!) * 4 ) = ceil( log2(6) * 4 ) = ceil(10.33984) = 11
g^33 = 3 * n_min = 3 * 11 = 33
```

This derivation is combinatorial and heuristic; see Remark 2.3 for the precise mathematical status. Below 33 cycles the system is conjectured to be fragile. At or above 33, the organizing invariant is conjectured to persist across disruption.

Definition 3.3 -- Circuit Saturation at g^64 = 64.

After 64 cycles the system maps its entire possibility space (Complete Completeness):

```
x_64 = G^64(x_0)
```

Definition 3.4 -- The Spiral Return.

The return is a new application of G at circuit scale on the contact manifold:

```
x_0 ->^{G^64} x_64 ->^{G^64} x_0' with x_0' != x_0
```

This is a spiral, not a loop. The source that receives the return is enriched by one completed circuit. The action variable z accumulates dissipation along the return: dz/dt = f(x) >= 0 on Gamma, encoding the thermodynamic cost of the full circuit in the contact form alpha = dz - lambda.

4. Theorem T1 -- The Generative Time Circuit

Theorem T1 (Generative Time Circuit Theorem -- GTCT).

> In the TOGT operator algebra on the contact manifold M = S x R with contact form alpha = dz - lambda, time is the generative circuit operator T: the fifth stage of the operator chain C -> K -> F -> U -> T. For any dm3 system D = (S, g, Y, Gamma, V, sigma) that has completed at least g^33 = 33 generative cycles, the spiral return x_0' = G^64(G^64(x_0)) satisfies x_0' != x_0. The action variable z increases monotonically along the return, encoding the thermodynamic cost of the completed circuit. Structural stability holds within the dm3 radius epsilon_0 = 1/3 (Theorem D, Volume II).

5. Complete Proof

The proof proceeds in five steps using only the operator algebra and the dm3 contact-geometric structure established in Volumes I and II.

Step 1. Circuit geometry -- not a line.

The contact manifold M = S x R carries the contact form alpha = dz - lambda. Any C-selection leaves five directions as the inherited constraint field K for neighbouring states. Space is a record of operator interactions. The operator chain is explicitly a circuit: C -> K -> F -> U -> T -> source. Therefore, time T is the full circuit operator, not a linear parameter. (QED Step 1)

Step 2. Stability via the g^33 conjecture (the Cajueiro Principle).

By Definition 3.2, the heuristic stability index 33 is the conjectured minimum number of cycles for the three invariants to achieve simultaneous, robust closure. The dm3 Lyapunov function V(rho) = (1/2)(rho - 1)^2 guarantees that for any perturbation ||Y|| <= epsilon < epsilon_0 = 1/3, the transverse deviation xi(t) = rho(t) - 1 satisfies the Gronwall bound (Theorem D, Volume II):

```
|xi(t)| <= |xi(0)| * exp( (|mu_max| + 3*epsilon) * t ) -> 0 when epsilon < 1/3
```

The system is self-sustaining and regenerates after any disruption within epsilon_0. (QED Step 2)

Step 3. Saturation at g^64 = 64.

In the dm3 contact system on M = S^1 x R, the flow of G over T* = 2*pi is governed by the contact normal form (Theorem C, Volume II):

```
d(rho)/dt = mu_max * (1 - exp(-beta*z)) * rho
d(theta)/dt = omega
dz/dt = omega - |mu_max| * rho^2 * exp(-beta*z)
```

The transverse multiplier is lambda_perp = exp(mu_max * T*) = exp(-4*pi) ~ 3.49 x 10^(-6) in the canonical toy model (mu_max = -2, T* = 2*pi). After 64 applications, the full possibility space is mapped: x_64 := G^64(x_0). (QED Step 3)

Step 4. Spiral Return.

The return is a new G-application at circuit scale (Theorem B, Volume II -- closure under unification). In contact coordinates, the return accumulates dissipation in the action variable z:

```
x_0 ->^{G^64} x_64 ->^{G^64} x_0' with x_0' != x_0
```

The Fold operator F curves the trajectory back onto itself (first self-reference). The action variable z satisfies dz/dt >= 0 along Gamma, so the source state x_0' carries a strictly greater accumulated entropy than x_0. Macroscopic linear history remains irreversible (entropy and decoherence above the manifold); at the generative level the source is enriched -- not rewritten -- by one completed circuit. (QED Step 4)

Step 5. Structural stability of the return.

By Theorem D (Volume II), all operators preserve the dm3 structure within the stability radius:

```
epsilon_0 = |mu_max| / [2 * (1 + sup_Gamma ||Hess V||)] = 2 / [2*(1+2)] = 1/3
```

Small perturbations to the operator chain within epsilon_0 = 1/3 do not destroy the enriched return x_0'. The spiral return is a generic property of any dm3 system operating above the g^33 stability index, not an artifact of fine-tuned parameters. (QED Step 5)

Conclusion. Once the g^33 stability index is reached and the full circuit closes at g^64, the action variable z accumulates strictly positive dissipation over every complete circuit. The source state is therefore enriched -- carrying the accumulated thermodynamic cost of one full circuit -- rather than returned to its original configuration. This is the spiral return. Structural stability holds within epsilon_0 = 1/3. (QED Theorem T1)

6. The Gronwall Inequality and epsilon_0 = 1/3

Theorem (Gronwall Inequality -- Differential Form).

Let u(t) >= 0 satisfy u'(t) <= k(t)*u(t) + f(t), u(0) = a >= 0, with k, f continuous. Then:

```
u(t) <= a * exp( int_0^t k(s) ds ) + int_0^t f(s) * exp( int_s^t k(r) dr ) ds
```

Proof sketch. Define mu(t) = exp(-int_0^t k ds). Then d/dt[u * mu] <= f * mu. Integrate and divide by mu(t). (QED)

Application to the dm3 Transverse Deviation.

The transverse deviation xi(t) = rho(t) - 1 from Gamma = {rho = 1} under perturbation ||Y|| <= epsilon satisfies:

```
|d(xi)/dt| <= |mu_max| * |xi(t)| + epsilon * (1 + sup_Gamma ||Hess V||) *
|xi(t)|
```

By Gronwall with k = |mu_max| = 2 and sup ||Hess V|| = 2:

```
epsilon_0 = 2 / [2 * (1 + 2)] = 1/3 (exact)
```

epsilon	Exponent at t=1	|xi(1)| / |xi(0)|	Decay?	Status
0.20 (< 1/3)	-2 + 0.6 = -1.4	exp(-1.4) ~ 0.247	Yes	Stable
0.333 (= 1/3)	-2 + 1.0 = -1.0	exp(-1.0) ~ 0.368	Yes	Marginal
0.40 (> 1/3)	-2 + 1.2 = -0.8	exp(-0.8) ~ 0.449	No	Slower decay
0.80 (> 1/3)	-2 + 2.4 = +0.4	exp(+0.4) ~ 1.492	No	GROWING

Table 1: Numerical verification of epsilon_0 = 1/3.

7. Falsifiability

Three independent falsifiability conditions:

F1. Absence of spiral enrichment (x_0' = x_0, no accumulated z) in dm3 contact simulations above the g^33 stability index would refute T1.

F2. Violation of dm3 structural stability (epsilon_0 = 1/3) under operator perturbations within the stated bounds would refute T1.

F3. A formal proof in Lean 4 (AXLE repository) that the Gronwall bound does not imply transverse contraction toward Gamma would refute T1.

8. Research Directions

The contact-geometric results of Sections 5-6 are classical dissipative dynamics. The connections described below are structural analogies and open research problems, not theorems. No retrocausal claims are made in this paper.

8.1 Operator Grammar and Time-Symmetric Quantum Mechanics.

The two-state vector formalism (TSVF) of Aharonov and Vaidman [1] assigns both a forward-evolving and a backward-evolving state to each quantum system. The operator chain G = U * F * K * C has a structural resemblance to this bidirectional picture: the Fold operator F introduces self-reference at mid-circuit, and the Unfold operator U projects outward. Whether this resemblance can be made precise -- by constructing a map from the dm3 contact system to the TSVF Hilbert-space formalism -- is an open problem. No retrocausal claim is made in this paper.

Open Problem 8.1. Construct, if possible, a functor from the dm3 operator category to the category of pre- and post-selected quantum states in the TSVF, preserving the stability radius epsilon_0 = 1/3.

8.2 Delayed-Choice Experiments and the Fold Operator.

In the delayed-choice quantum eraser (Kim et al. [2]), a future measurement choice determines which past path information is available. The Fold operator F -- which introduces the first self-reference in the chain -- is structurally analogous to the beam-splitter choice in that experiment: it is the point at which the system curves back onto its own prior state. This analogy motivates the following research direction but does not constitute a proof of any physical connection.

Open Problem 8.2. Identify an experimental observable in a quantum optics or NMR system that would distinguish spiral enrichment (x_0' != x_0 after circuit closure) from linear return (x_0' = x_0), and determine whether the dm3 stability index g^33 has a measurable counterpart in that system.

8.3 Entanglement Swapping and the Spiral Return.

Ma et al. [3] demonstrated entanglement swapping between photons that never coexisted, suggesting non-trivial temporal structure at the quantum scale. The spiral return x_0 -> x_64 -> x_0' with x_0' != x_0 in the dm3 circuit is formally analogous: the source state is enriched by one completed circuit without returning to its original configuration. Whether this formal analogy corresponds to a physical mechanism is unknown.

Open Problem 8.3. Determine whether the contact-geometric spiral return, formalized via the Floquet multiplier lambda_perp = exp(-4*pi), admits a quantum information interpretation consistent with the entanglement swapping results of Ma et al. [3].

8.4 Negative Group Delay and the Weave Scale.

Angulo et al. [4] report experimental evidence of negative group delay -- a well-understood phenomenon in classical wave optics (anomalous dispersion) that does not require retrocausality. The dm3 framework does not explain or predict negative group delay. The reference is included as context for the broader research landscape, not as evidence for the operator grammar.

8.5 Formal Verification in AXLE (Lean 4).

The AXLE repository (github.com/TOTOGT/AXLE) contains Lean 4 proof obligations for the dm3 framework. The highest-priority open items relevant to this paper are:

(a) Formal verification of the Gronwall bound and epsilon_0 = 1/3 (Section 6) -- currently the most tractable target.

(b) Formalization of the g-series taxonomy (Definition 2.2) as a Lean 4 inductive type, with the index-33 conjecture stated as a *sorry* to be discharged by future work.

(c) The Poincare-to-Collatz correspondence -- the hardest open item, currently marked as a *sorry* in AXLE -- is not required for the results of this paper.

9. Connection to Established Frameworks

Framework	What it provides	GTCT research direction
Two-State Vector Formalism (Aharonov & Vaidman [1])	Forward + backward evolving states; future measurements constrain past correlations.	Open Problem 8.1: functor from dm3 operator category to TSVF, preser
Delayed-Choice Quantum Eraser (Kim et al. [2])	Future measurement retroactively determines past path/wave behavior	Open Problem 8.2: identify observable distinguishing spiral from linear r
Entanglement Swapping (Ma et al. [3])	Photons that never coexisted show correlations; non-local temporal structure at quantum scale.	Open Problem 8.3: quantum information interpretation of lambda_perp =
Negative Group Delay (Angulo et al. [4])	Photons appear to exit medium before entering; classical anomalous dispersion	Noted as research context only; dm3 makes no prediction about this phe
Contact Hamiltonian Mechanics (Herczeg & Waldron [5])	Generally covariant QM on phase-spacetime using the contact structure.	dm3 is the dissipative realization with explicit limit-cycle attractors and st

Table 2: GTCT research directions in relation to established frameworks.

10. Exercises

Six graded exercises. A student who completes all six has verified the theorem from multiple independent angles.

E1 [Introductory]. Verify numerically that log2(3!) * 4 = 10.3398... and therefore ceil(10.3398) = 11. Then verify 3 * 11 = 33. Explain in your own words why multiplying by 3 rather than 1 is necessary for the triple-confirmation robustness criterion in Remark 2.3.

E2 [Calculus]. Prove the Gronwall inequality in differential form using the integrating factor mu(t) = exp(-int_0^t k(s) ds). Apply it to the dm3 transverse deviation xi(t) with k = |mu_max| = 2 and show epsilon_0 = 1/3 is exact.

E3 [Differential Equations]. Solve the dm3 normal form equations numerically (Euler method, dt = 0.01) for (rho_0, theta_0, z_0) = (1.2, 0, 0) with mu_max = -2, beta = 1, omega = 1. Integrate for 10 periods. Plot rho(t) and show convergence to Gamma = {rho = 1}. Compute lambda_perp = exp(-4*pi) and compare to the numerical rate.

E4 [Contact Geometry]. Show that the dm3 contact form alpha = dz - lambda is non-degenerate: alpha ^ (d(alpha))^n != 0 everywhere. Explain why symplectic geometry (d(lambda) only, no z variable) cannot support attractors on compact manifolds but contact geometry can.

E5 [Quantum Foundations]. Read Kim et al. (2000, PRL 84:1). Identify which of C, K, F, U corresponds to each stage of the delayed-choice quantum eraser experiment. Which operator is the beam-splitter choice? Which is the Fold that makes the future measurement retroactively constrain the past path? Write a 400-word analysis.

E6 [Research Level]. The g^33 stability index conjectures that spiral enrichment requires at least 33 operator cycles. Design an experimental protocol (quantum optics or NMR) to test this conjecture. Specify: (1) system and rationale; (2) operationalization of 33 cycles; (3) observable distinguishing spiral from linear return; (4) result that would falsify the conjecture. Submit as a 1-page research proposal.

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C -> K -> F -> U -> T -> source

Dedicated to Vic, Alice, and Sarah -- once tiny, always strong.